

8. MERITS, LIMITATIONS AND PITFALLS OF DATA COLLECTION AND ANALYSIS

The pros, cons, and drawbacks of the various data collection and analysis methods employed in this study must be discussed in detail in this section. This section includes a complete per-strand analysis and a cross-strand synthesis assessment.

8.1 QUANTITATIVE SURVEY FULL ANALYSIS

MERITS

Merit 1 Multi-sector diversity: 8 sectors are represented, such as Energy, Finance, Healthcare and Defence. This breadth allows findings to be not specific to a single CI context and can be generalized across the RGS client base. An original contribution is the absence of a similar multi-sector primary survey on AI-IDS in CI in literature.

Merit 2 Likert scale is suitable for quantitative comparison: Six Likert items on consistent 1–5 scales can be used for descriptive statistical analysis (means, medians, modes, SDs) and can be compared directly across dimensions.

For example, Q4 mean 2.32 can be directly compared to Q9 mean 3.50a finding only possible with Likert data.

Merit 3 Multi-select Q7 shows barrier co-occurrence: Multi-select Q7 format offers more data on barriers than single-choice. Skills gaps and costs co-occur in the results, and RGS will need to tackle both at once.

Merit 4 Q15 open text adds qualitative triangulation: Five open text responses offer within-survey qualitative data that supports Likert findings – the concept drift mention independently supports the simulation finding in Section 5.2.

LIMITATIONS

Limitation 1 Sample size ($n=22$): $n=22$ is less than the number required for parametric inference ($n \geq 30$). All statistics are only descriptive. There are no inferential statistics (t-tests, correlations) reported. No claims of statistical significance are made.

Limitation 2 Early-career skew (72.7% with 0-2 years): The most significant limitation. More advanced practitioners would offer more context for IDS deployments. Likert implies that it might be influenced by the extent of operational exposure a practitioner with 15 years of ICS security experience might score higher on adversarial concern than a practitioner with 12 months of general IT experience.

Limitation 3Purposive sampling and self-selection: Respondents accessed via LinkedIn may have more positive attitudes towards AI than the general CI security population, potentially overstating the 40.9% support score.

Limitation 4Q10 categorical instead of Likert: This precluded mean calculation for XAI trust impact, which limits comparison with the six Likert dimensions. In the future, a 5-point scale should be used.

PITFALLS

Pitfall 1Bimodal Q6 distribution misread as neutral: The Q6 mean (2.91) is in the middle, but there is a bimodal distribution with Mode=1. Only reporting the mean would represent this finding as moderate ambivalence. This pitfall was avoided by reporting the full distribution as well as the mean in each section.

Pitfall 2: Hawthorne effect and survey framing A survey title that is AI-focused can lead to more AI-positive responses, creating a frame-induced bias. This is not completely removed, but recognized in all sections where Q6 support scores are mentioned.

MITIGATION STRATEGIES APPLIED

- Sample limitations explained and emphasized in each survey section. Means always provided with full distributions, modes, and SDs to avoid misinterpretation of bimodal patterns.
- No claim of inferential statistics.
- All survey results are checked against the simulation and literature before a conclusion is made.
- Q15 qualitative data to validate and enrich Likert findings.

8.2 ALGORITHM SIMULATIONFULL ANALYSIS

MERITS

Merit 1Controlled, reproducible experimental conditions: All eight algorithms are tested under identical conditions, so as to remove confounding variables that exist in CI deployments in the wild. Documented simulation parameters guarantee the reproducibility.

Merit 2OT-specific datasets (SWaT, CIC-IDS): SWaT is the only publicly available dataset from a real ICS water treatment testbed. The majority of the published AI-IDS studies are based on enterprise IT datasets. This study's OT data use fills the explicit gap identified in Gupta (2023) and Darktrace (2024).

Merit 3Latency benchmarking provides operational relevance: Latency measurement in addition to F1 provides practical deployment relevance that most academic studies lack. The 38ms

LSTM finding, which falls in the 100–200ms OT range, is a deploy ability statement that addresses practitioner concerns directly.

Merit 4 Novel concept drift and adversarial analysis: The drift simulation for 30 weeks and the adversarial analysis for 10 types are original concepts, with no direct equivalent in the literature. The quantification of 29.4% F1 drift degradation and 63% PGD-induced collapse gives RGS new, specific risk parameters.

Merit 5 SMOTE application: SMOTE application is explicitly documented with its impact on F1 score (0.12-0.19 improvement) which increases the validity of the results of benchmarking and prevents artificially inflated accuracy due to prediction by majority class.

LIMITATIONS

Limitation 1 Digital twin fidelity gap: Real OT networks are filled with proprietary protocols, physical process dependencies, and operational anomalies that are hard to simulate. Environmental noise factors can result in 10-15% lower F1 scores on live deployments.

Limitation 2 Discrete drift events only: Ten individual drift events are not representative of the continuous drift events that are typical of real CI environments. Degradation numbers may be underestimates for compound drift situations.

Limitation 3 Individual adversarial attack testing: Real adversarial actors use compound strategies simultaneously. The combined evasion rate is probably lower than individual attack testing.

PITFALLS

Pitfall 1: Dataset-specific overfitting (SWaT and CIC-IDS have known attack signatures). LSTM Autoencoders might have captured anomaly patterns in the dataset. Generalizability would be enhanced by validation with a third, independent data set.

MITIGATION STRATEGIES APPLIED

- Conservative 10–15% performance discount applied to live deployment projections.
- Compound adversarial scenarios identified as priority future research.
- Performance claimed to be 'simulation-validated' all the way through, rather than 'deployment-proven'.

- All supervised algorithms were five-fold cross validated to minimize overfitting.

8.3 LITERATURE REVIEWFULL ANALYSIS

MERITS

Merit 1Systematic source selection: 10 sources selected against explicit inclusion criteria (within 5 years, peer-reviewed/institutional, directly relevant to at least one RQ). Minimizes selection bias as opposed to ad hoc citation.

Merit 2Structured gap analysis: Table 3 clearly illustrates what each source addresses and how this study contributes to the gap. This shows original work and not copying.

LIMITATIONS

Limitation 1: Enterprise IT bias: Most of the published research on AI-IDS is in enterprise IT environments. Validated benchmarks are scarce and are specific to the OT which requires the simulation strand.

Limitation 2AI security is a rapidly changing field. The sources from 2022-2023 may be underestimating current AI capabilities or adversarial risks.

PITFALLS

Pitfall 1Uncritical citation: Statistics were not cited without critically assessing the methodology used to produce them, but methodological gaps in each source were identified (Table 3).

8.4 CROSS-STRAND SYNTHESIS: METHODOLOGICAL COHERENCE

The three strands all lead to the same five conclusions: (1) the current IDS is insufficient; (2) AI anomaly detection fills this gap but has adoption challenges; (3) there is a broad appreciation of XAI; (4) FL awareness is greater than willingness to use due to governance gaps; (5) adversarial risks are not well understood. These conclusions are supported by the convergence of the independent strands, even though there are limitations.

The one area of partial non-convergence, adversarial concern (Q13 mean 2.82/5 vs PGD 63% F1 collapse), is not a methodological inconsistencies but a real knowledge gap, an important finding which is directly motivating the mandatory adversarial testing in Recommendation 1.